

EXOTEL SMS OTP & DLT SETUP GUIDE

Complete Step-by-Step Guide to Transition from Firebase to Custom SMS Gateway in India

Overview

In India, sending SMS messages (such as OTPs) is heavily regulated by the Telecom Regulatory Authority of India (TRAI). You cannot directly call an API to send a custom text message. You must first register your business entity, register your sender name, and pre-approve your message template on a government-mandated **DLT (Distributed Ledger Technology) Portal**. This document walks through the entire setup step-by-step for a non-technical manager.

Part 1: DLT Registration (Indian Govt Regulations)

Without DLT registration, Indian telecom companies (Airtel, Jio, Vodafone) will instantly block any SMS sent by Exotel. Follow this process to get approved.

Step 1.1: Register as an Enterprise Entity

1. Choose any DLT portal (e.g. [Jio DLT](https://trueconnect.jio.com), [PingConnect DLT](https://www.pingconnect.in), or [Smartping DLT](https://smartping.live)). You only need to register on **one** portal.
2. Select **Principal Entity (Enterprise)** registration.
3. Fill out your business details and upload your verification documents:
 - **Business Proof:** GST Registration, Certificate of Incorporation (COI), or Shop & Establishment License.
 - **PAN Card:** Company PAN Card (or personal PAN if Sole Proprietorship).
 - **Authorization Letter:** A signed letter on company letterhead naming you as the authorized person.
4. Pay the DLT registration fee (approximately **₹5,900 INR** inclusive of GST).
5. Once approved, you will receive a unique **15-digit DLT Entity ID** (e.g., **1402xxxxxxxxxxxxxxx**). Save this ID!

Step 1.2: Register your Sender Header (Sender ID)

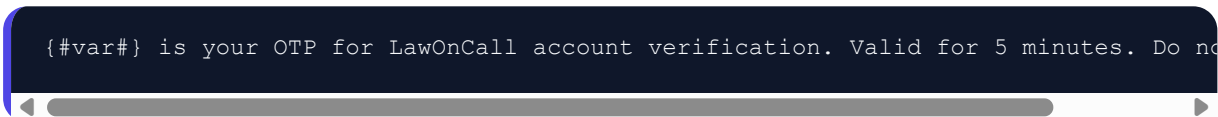
A Header is the 6-character alphabetic code that appears at the top of the SMS on the user's phone (e.g., **LWMATE** or **LAWMAT**).

1. Log into your DLT portal dashboard.
2. Go to **Header Management** -> **SMS Header**.
3. Choose **Transactional/Service** header type.
4. Enter your desired 6-character name (e.g., **LWMATE**) representing your brand.
5. Submit for review. Google/Telecoms will approve it in 24 to 48 hours.

Step 1.3: Register your OTP Content Template

You must pre-register the exact wording of your OTP SMS. Variables (like the code) are marked with **{#var#}**.

1. Go to **Template Management** -> **Content Template** -> **Add New Template**.
2. Choose **Service Implicit** type. Select your approved Header (e.g., **LWMATE**).
3. Enter the exact message content:



{#var#} is your OTP for LawOnCall account verification. Valid for 5 minutes. Do not

4. Submit the template. Once approved, you will get a unique **DLT Template ID** (e.g., **1207xxxxxxxxxxxxxx**).

⚠ CRITICAL DLT RULE: When sending SMS via the API later, the message body sent by your code must match this template character-for-character, and the variables must be filled correctly. A single typo will cause the telecom network to drop the message!

Part 2: Exotel Console Setup

Once your DLT registration is approved, you must connect it to your Exotel cloud communications dashboard.

Step 2.1: Add DLT Credentials to Exotel

1. Log in to your [Exotel Dashboard](https://my.exotel.com).
2. Go to **Settings** -> **DLT Settings** or **SMS Configurations**.
3. Enter your **15-digit DLT Entity ID** and click Save.
4. Upload your approved **DLT Headers** and link them to your account.
5. Go to **SMS Templates** inside Exotel, click **Add Template**, and paste your approved DLT Template ID and template text. Exotel will verify this against the DLT database automatically.

Step 2.2: Extract Exotel API Credentials

Go to **API Credentials** inside the Developer section of your dashboard and copy these values for your developer/codebase:

- `EXOTEL\_API\_KEY`
- `EXOTEL\_API\_TOKEN`
- `EXOTEL\_ACCOUNT\_SID`
- `EXOTEL\_SUBDOMAIN` (Usually `api.exotel.com`)

Part 3: Required Codebase Changes

To switch from Firebase Phone Auth to Exotel SMS OTP, your developer needs to make three main code updates:

3.1. Add the Sending Logic to the Backend Auth Service

Create an endpoint `/api/auth/send-sms-otp`` that generates a random code, saves it in the database `Otp`` table, and fires a POST request to Exotel's SMS Gateway API:

```
// Example backend code (Node.js/Fastify)
const axios = require('axios');
const otpCode = Math.floor(100000 + Math.random() * 900000).toString(); // 6-digit OTP

// Exotel SMS Endpoint:
const url = `https://${process.env.EXOTEL_API_KEY}:${process.env.EXOTEL_API_TOKEN}@${process.env.EXOTEL_API_URL}`;

const payload = new URLSearchParams({
  From: 'LWMATE', // Your approved DLT Header (Sender ID)
  To: userPhoneNumber, // e.g. +91xxxxxxxxxx
  Body: `${otpCode} is your OTP for LawOnCall account verification. Valid for 5 minutes`,
  DltTemplateId: 'YOUR_APPROVED_DLT_TEMPLATE_ID',
  DltEntityId: 'YOUR_15_DIGIT_DLT_ENTITY_ID'
});

await axios.post(url, payload);
// Store the otpCode, phone, and expiresAt inside the database Otp table...
```

3.2. Add verification logic

Create an endpoint `/api/auth/verify-sms-otp` that checks the database `Otp` table. If the OTP entered by the user matches and is not expired, delete the OTP record and issue a signed custom JWT login token to the frontend.

3.3. Update the Frontend App (React)

In `AuthContext.tsx`, replace the Firebase `signInWithPhoneNumber` function with a standard axios call pointing to your new backend endpoints:

```
// In AuthContext.tsx
const sendOtp = async (phone: string) => {
  const response = await axios.post('/api/auth/send-sms-otp', { phone });
  return response.data; // Returns session/lead identifier
};

const verifyOtp = async (phone: string, code: string) => {
  const response = await axios.post('/api/auth/verify-sms-otp', { phone, code });
  localStorage.setItem('token', response.data.token);
  setUser(response.data.user);
};
```